Roll No. | | | | | | | | | | | | Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-4)

# PHARMACEUTICAL ORGANIC CHEMISTRY-III

Subject Code : BP-401T

M.Code : 75843

Date of Examination : 02-01-2024

Time : 3 Hrs. Max. Marks : 75

**INSTRUCTIONS TO CANDIDATES :**

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

## SECTION-A

**1. Write short notes on :**

a) What is the key difference between a stereoisomer and a structural isomer?

b) Define enantiomerism in the context of stereochemistry.

c) How does a chiral molecule differ from an achiral molecule?

d) What is meant by the term "racemic modification" in stereochemistry?

e) Differentiate between the terms "cis" and "trans" when used to describe geometrical isomers.

f) Describe the conformational isomerism in ethane and explain how it arises.

g) What are the key differences between pyrazole and pyridine in terms of their ring structure?

h) What is a key structural feature of azepines and how does it influence their reactivity?

i) In the Birch reduction, what type of compounds are typically reduced to yield dienes?

j) What is the product of a Clemmensen reduction of a carbonyl compound?

## SECTION-B

2. Describe the DL system of nomenclature for optical isomers. What are its limitations?

3. Describe the Oppenauer oxidation and Dakin reaction, including its mechanism.

4. Explain the concept of aromaticity and describe the relative aromaticity of pyrrole, furan and thiophene.

## SECTION-C

5. Discuss the role of elements of symmetry in determining whether a molecule is chiral or achiral.

6. Discuss the conditions under which biphenyl compounds exhibit optical activity.

7. Explain the differences between stereospecific and stereoselective reactions.

8. Discuss the basicity of pyridine, explaining why it is a weaker base compared to typical amines.

9. Compare and contrast the structural features and reactivity of indole, quinoline and isoquinoline.

10. Elaborate on the Wolff-Kishner reduction and its utility in converting carbonyl compounds.

11. Compare and contrast the mechanisms of metal hydride reductions using $NaBH_4$ and $LiAlH_4$.

12. What are the chair and boat conformations in cyclohexane? Why is the chair conformation more stable?

13. Explain the concept of asymmetric synthesis in the context of chiral molecules.

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**NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.**



DEC-2023